ACADEMIC CENTRAL

PRACTICE EXAMINATION

GRADE 11

AGRICULTURAL MANAGEMENT PRACTICES

PAPER 1

MARKS: 100 **TIME: 2 hours**

Topics: animal breeding and feeding, animal health and welfare, crop production, farm business and marketing.

Written by Academic Central for practice, following the CAPS curriculum for Grade 11. The farms and people are invented. It is not a Department of Basic Education examination paper.

INSTRUCTIONS AND INFORMATION

1. Answer ALL the questions. Show ALL calculations. Round off to TWO decimal places where necessary.

QUESTION 1

1.1 Choose the correct answer. Write only the letter (A–D).

1.1.1 The gestation period of a sow is about …

A 21 days. B 114 days.
C 150 days. D 283 days.

1.1.2 The Pearson square is used to …

A calculate feed rations. B measure rainfall.
C identify animals. D test soil pH.

1.1.3 Which cattle disease is spread by ticks?

A Redwater B Mastitis
C Newcastle disease D Bloat

1.1.4 Growing two or more crops in alternate rows in the same field at the same time is …

A intercropping. B monocropping.
C crop rotation. D fallowing.

1.1.5 Gross margin is calculated as …

A gross income − variable costs. B gross income − fixed costs.
C gross income + total costs. D fixed costs + variable costs.

(5 × 2)
(10)

1.2 Give ONE term for each description.

1.2.1 Mating animals of two different breeds

1.2.2 Placing semen into the reproductive tract of a female artificially

1.2.3 The amount of water a soil holds after excess water has drained away

1.2.4 Increasing the value of a product by processing or packaging it

1.2.5 An organisation of farmers who buy inputs and sell products together (5 × 2)
(10)

1.3 Indicate whether each statement is TRUE or FALSE. If FALSE, correct the underlined word(s).

1.3.1 Colostrum, the first milk after birth, contains <u>antibodies</u> that protect the newborn.

1.3.2 Castration makes bull calves <u>more aggressive</u>.

1.3.3 <u>Fixed</u> costs change with the size of production.

1.3.4 A <u>tensiometer</u> measures soil moisture.

1.3.5 Inbreeding increases hybrid vigour. (5 × 2) (10)

[30]

QUESTION 2: ANIMAL PRODUCTION

2.1 Define *crossbreeding* and give ONE advantage of it. (3)

2.2 Write down the approximate gestation period of a cow, a ewe and a sow. (3)

2.3 A cow was mated on 1 March. Using a gestation period of 283 days, calculate the expected calving date. Show your working. (3)

2.4 A farmer wants 1 000 kg of pig feed with 16% crude protein (CP), made from yellow maize (9% CP) and soybean oilcake (44% CP). Use the Pearson square to calculate the mass of each ingredient. (6)

2.5 Name ONE notifiable disease of cattle and explain why such diseases must be reported to the state veterinarian. (3)

2.6 Name THREE of the "five freedoms" of animal welfare. (3)

2.7 Name TWO methods of permanently identifying livestock. (2)

2.8 Why should a farmer follow a vaccination programme? (2)

[25]

QUESTION 3: CROP PRODUCTION

3.1 What is irrigation scheduling? Name TWO methods a farmer can use to decide when to irrigate. (4)

3.2 A 3 ha maize field needs 120 kg of nitrogen (N) per hectare. The farmer uses LAN, which contains 28% N. Calculate the total mass of LAN needed. (5)

3.3 Explain integrated pest management (IPM) and name THREE methods it combines. (5)

3.4 Give THREE signs that maize is ready to harvest for storage. (3)

3.5 Give TWO causes of post-harvest losses and a way to prevent each. (4)

3.6 Give FOUR advantages of producing vegetables in a greenhouse compared with an open field. (4)

[25]

QUESTION 4: FARM BUSINESS AND MARKETING

Mpho raised 1 000 broilers. Each bird was sold at 2 kg live mass for R38 per kg. Her variable costs were: chicks R9 000; feed R38 000; veterinary costs R2 500; electricity R3 000; labour R6 000.

4.1 Define *gross margin*. (2)

4.2 Calculate: (a) the gross income; (b) the total variable costs; (c) the gross margin; (d) the gross margin per bird. (8)

4.3 Give TWO advantages for small-scale farmers of joining a cooperative. (4)

4.4 Give ONE example of value adding for Mpho's broilers and explain how it could increase her income. (3)

4.5 Name ONE South African law that protects animals and give ONE thing it requires of farmers. (3)

[20]

TOTAL: 100

END OF QUESTION PAPER